Venkata Lakshmi Parimala Pasupuleti

Senior Software Engineer · AI Platform · Python · LLM APIs · RAG · MCP · Observability · AWS · Security & Compliance · 9yr Production

+1-703-261-5410 | lakshmiparimalap@gmail.com | linkedin.com/in/venkata-lakshmi-parimala-pasupuleti |
github.com/venkatalakshmiparimala | Virginia, USA (Remote)

PROFESSIONAL SUMMARY

Senior software engineer with 9+ years of production experience building AI platforms, LLM integrations, RAG infrastructure, and internal tooling that other engineers build on. AgentProbe is the direct proof point for this role: an internal AI platform with multi-provider LLM routing (Anthropic API, OpenAI API, Ollama), RAG infrastructure (ChromaDB vector database, chunking, embedding optimization), MCP integration for unified AI-to-system data retrieval, 5-metric observability engine (correctness, groundedness, hallucination rate, latency, token efficiency), and a CI-integrated deployment gate - exactly the kind of reusable platform Axon's Cortex team builds and operates for internal users. Leasel completes the picture: a production agentic system with 8-agent LangGraph orchestration, tool use, HITL safety gates, and deterministic guardrails - the agent workflow and safe tool use patterns the Corporate AI team defines as standards for other engineers to follow. My security and compliance background is deep: 7 years of PHI/HIPAA-compliant systems with Libsodium-encrypted pipelines, role-based access controls, audit logging, and zero security incidents - the secure-by-default API design and audit requirements the Cortex platform needs for enterprise system integrations. I leverage Claude Code daily for AI-assisted development, and apply prompt engineering, context engineering, and RAG optimization as first-class engineering practices. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

TECHNICAL SKILLS

AI Platform & LLM Integration: Python (9yr), TypeScript, JavaScript, LangGraph, LangChain, Agentic Workflows, Multi-Agent Orchestration, LLM APIs (Anthropic, OpenAI, Ollama), MCP (Model Context Protocol), Prompt Engineering, Context Engineering, Tool Use, HITL, Guardrails, RAG Pipelines, ChromaDB, Vector Databases, Embeddings, Chunking Strategies, Semantic Search, NLP, BERT, spaCy, Claude Code

Observability & LLMOps: LLM Evaluation, LLM-as-Judge, Hallucination Detection, Observability & Metrics Pipelines, Model Monitoring, Execution Tracing, Decision Logging, Cost Monitoring, Dashboards, CI/CD Regression Gate, GitHub Actions CI/CD, Automated Testing, FastAPI, REST APIs, PyTorch, scikit-learn

Cloud, Security & Infrastructure: AWS (S3, EC2, RDS, Glue), Docker, Microservices, RabbitMQ (event-driven async), Redis, PostgreSQL, MySQL, Oracle, Security & Encryption (Libsodium), PHI/HIPAA Compliance, Audit Logging, RBAC, Secrets Management, Performance Engineering, Linux, Git, Agile/Scrum, React.js

PROJECTS

AgentProbe: Internal AI Platform - LLM Routing, RAG, MCP & Observability

AI Platform · LLMOps · RAG · MCP · Observability

Python · FastAPI · LangGraph · ChromaDB · MCP · GitHub Actions · Anthropic API · OpenAI API · Ollama | github.com/venkatalakshmiparimala/AgentProbe

Production internal AI platform - reusable infrastructure for LLM routing, RAG, tool integration, and observability that other engineers build on. **LLM platform:** multi-provider routing (Anthropic API, OpenAI API, Ollama) via clean API abstractions; MCP integration for unified AI-to-system data retrieval across providers. **RAG infrastructure:** ChromaDB vector database, chunking strategy optimization, embedding selection, semantic search. **Observability:** 5-metric instrumentation engine (correctness, groundedness, hallucination rate, latency, token efficiency); execution tracing; CI regression gate blocking quality degradation before deployment. **Security:** audit logging for every LLM call; access controls; secrets management patterns. Zero regressions across 1,240+ test cases. Open-source, enterprise-deployable, fully documented with starter kit patterns.

Leasel: Production Agentic System - Tool Use, HITL & Safe Guardrails

Agentic AI · Tool Use · HITL · Security · Audit Logging

Python · LangGraph (8-agent) · FastAPI · OCR · SQLite · Claude Code | github.com/venkatalakshmiparimala/Lease_complaince_rag

Production agentic AI system demonstrating safe tool use, multi-agent coordination, and human-in-the-loop controls - the agent design patterns and guardrail standards a platform team defines for other engineers to follow. **Agent orchestration:** 8-agent LangGraph StateGraph with conditional routing, dynamic task delegation, and tool invocation across document retrieval, structured data, configuration, and historical incident sources. **Safe tool use:** deterministic guardrails enforce that AI never resolves regulated decisions autonomously; HITL escalation gate on every low-confidence path. **Audit logging:** complete decision trail across all agents and tool calls; permissions respected throughout. 93.6% precision across 1,240 test cases.

AI on FHIR: Secure Enterprise API Integration - HIPAA & RBAC

Enterprise APIs · HIPAA · Security · REST · Full Stack

Python · Flask · React.js · TypeScript · Docker · BERT · spaCy · LLMs · FHIR REST APIs | github.com/venkatalakshmiparimala/AI_on_FHIR

Secure AI system integrating LLMs with enterprise REST APIs - translates natural-language queries into structured FHIR API calls; HIPAA-compliant RBAC with authentication, data isolation, and full audit logging; ~60% reduction in manual retrieval time; containerized with Docker. Demonstrates the secure enterprise system integration pattern (connecting AI to real organizational data while respecting permissions, auditing, and compliance requirements) the Cortex platform implements across Jira, Slack, M365, and Snowflake.

EXPERIENCE

Magizhs Tech Solution

May 2025 - Aug 2025

AI/ML Engineer - Production AI Platform Delivery

- Designed and deployed a production agentic AI system (LangGraph, FastAPI, Python) for an external customer - requirements gathering, architecture, implementation, automated testing with agent evaluation harness, and production deployment in weeks as sole technical owner. Documented patterns and runbooks for ongoing operation and future reuse.

George Mason University

Aug 2024 - May 2026

Graduate Teaching Assistant & Research Associate

- Built ML pipelines (PyTorch, scikit-learn, R^2=0.977); designed observability dashboards improving research comprehension 40%; mentored 600+ students in Python, SQL, and software engineering; translated complex technical outputs for non-technical stakeholders.

Entrata - Apartment & Affordable Housing SaaS

Sep 2018 - Aug 2024

Senior Software Engineer · Lehi, UT (Remote)

- **Secure, compliant production systems** - 100% PHI/HIPAA compliance with Libsodium-encrypted data pipelines; RBAC and access controls across multi-tenant SaaS; full audit logging for financial and compliance workflows; zero security incidents across 7yr of production operation; AWS S3/EC2/RDS/Glue; 99.5% uptime.
- **Platform engineering and observability** - built internal debugging and observability tools reducing triage time 40%; maintained bug count below 10 for 6+ consecutive months through proactive monitoring; Docker microservices (50% scalability improvement); RabbitMQ event-driven async with retry logic and idempotency; Redis caching; GitHub Actions CI/CD with automated testing throughout.
- **Enablement and best practices** - 300+ peer code reviews; mentored 5+ engineers on system design, security, and production practices; 120+ Agile sprints with product, compliance, and operations stakeholders; 20+ full-stack production modules (PHP, Python, PostgreSQL, TypeScript, React.js); leveraged Claude Code and AI-assisted development tools to accelerate engineering velocity.

Entrata India - Pune

Jan 2017 - Sep 2018

Associate Software Engineer → Trainee

- Built backend modules (PHP, Python, PostgreSQL) for production SaaS; data pipeline for 1.6M records; AWS S3 migration resolving 90% of legacy storage constraints.

EDUCATION

M.S. Data Analytics Engineering - George Mason University, Virginia — May 2026 · GPA 3.87

B.Tech Computer Science & Engineering - JNTU Anantapur — 2013-2017